B.J.Pol.S. 9, 219–236
Printed in Great Britain

Electoral Forecasting from Poll Data: The British Case

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Forecasting elections has long been regarded by political scientists as an interesting problem in its own right. But it assumes special importance for those countries that do not have fixed election dates. In Britain, for example, it is up to the prime minister to choose the date, within the statutory five year limit. Correct timing can clearly be crucial to the outcome, and the prime minister can be expected to go to considerable lengths to ensure that the election is called for the date most favourable to his party. But there lies the prime minister's problem: elections must be called three to four weeks before polling day. With what degree of accuracy can the result be forecast at the time the election is called?

A small but interesting literature on election forecasting has emerged in recent years. The forecasting techniques used in this literature vary widely, from crude extrapolation to sophisticated model building.[1] Up to now the emphasis has been on election night forecasting, in which the basic problem involves extracting the maximum amount of information from electoral returns, in order to forecast the outcome a few hours before it is finally known. For obvious reasons the techniques utilized in this context are of little use prior to election day.

The only reliable data available for purposes of election forecasting prior to polling day are opinion polls. Although the latest poll is often taken by politicians and the media as the best available indicator of an election outcome, and although poll data has been used for a variety of research purposes – ranging from the relationships between aggregate economic variables and political popularity, to the inferring of regime characteristics[2] – poll data has not been used much for purposes of forecasting, beyond simple projections into the future.

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[1] For an example of the latter, with a brief review of the existing literature, see P. Brown and C. Payne, 'Election Night Forecasting', *Journal of the Royal Statistical Society*, Series A, 138 (1975), 463–98. For an example of prediction not restricted to election nights see I. Budge and D. Farlie, *Voting and Party Competition* (London and New York: Wiley, 1977), Chap. 12 *passim*.

[2] For a review of this field see B. S. Frey and F. Schneider, 'On the Modelling of Politico-economic Interdependence', *European Journal of Political Research*, III (1975), 339–60. For Britain see C. Goodhart and R. Bhansali, 'Political Economy', *Political Studies*, XVIII (1970), 43–106 and W. Miller and M. Mackie, 'The Electoral Cycle and the Asymmetry of Government and Opposition Popularity: An Alternative Model of the Relationship between Economic Conditions and Political Popularity', *Political Studies*, XXI (1973), 263–79. For the United States see J. E. Mueller, 'Presidential Popularity from Truman to Johnson', *American Political Science Review*, LXIV (1970), 18–34; and J. A. Stimson, 'Public Support for American Presidents: A Cyclical Model', *Public Opinion Quarterly*, XL (1976), 1–11. For the Federal

0007-1234/79/2828-2010 $02.00

The purpose of this paper is to utilize British time-series poll data in order to develop efficient models for forecasting. The current observation of a poll contains information about the true state of public opinion, but it also contains measurement error and random noise. If the structural relationships between successive observations of a poll are identified and estimated it will usually improve the accuracy of the forecast compared with the simple projection of one poll result, since it will partly control these confounding errors.

The model-building strategy adopted in this paper was developed by Box and Jenkins, who have introduced a powerful set of models for representing time-series data.[3] Their model-building strategy involves identifying, estimating and forecasting from a model of a time series. They use stochastic models, and since these will be relatively unfamiliar to most political scientists the first section of the paper will be concerned with examining the nature of stochastic models in time-series analysis. The later sections are concerned with the estimation of and forecasting from such models, using poll data.

STOCHASTIC MODELS OF TIME-SERIES OBSERVATIONS

Box-Jenkins time-series analysis is concerned with the stochastic modelling of a sequence of observations that are assumed to be available at discrete, equispaced intervals. In such a model the observations evolve through time according to a probability process. One of the simplest examples of a stochastic process is that of a random walk. In this case successive changes in the observations are measured by a probability model, where the changes are assumed to be independent of each other.

To illustrate, if we denote the observation at time t as Z_t then a random walk could be modelled as follows:

$$Z_t - Z_{t-1} = a_t,$$

where

$$a_t = \begin{cases} +1 \text{ with a probability of } \tfrac{1}{2} \\ -1 \text{ with a probability of } \tfrac{1}{2}. \end{cases}$$

If the process starts at some time period Z_0, it will evolve as follows:

$$\begin{aligned} Z_1 &= Z_0 + a_1. \\ Z_2 &= Z_0 + a_1 + a_2, \\ &\vdots \\ Z_t &= Z_0 + a_1 + \ldots + a_t. \end{aligned}$$

This can be written

$$Z_t = Z_0 + a_t + a_{t-1} + a_{t-2} + \ldots + a_1. \tag{1}$$

Republic of Germany see C. Kirchgässner, 'Okonometrische Untersuchungen des Einflusses der Wirstschaftslage auf die Popularität der Partien', *Schweizerische Zeitschrift für Volkswirtschaft und Statistik*, CX (1974), 409–45.

[3] See G. Box and G. Jenkins, *Time Series Analysis: Forecasting and Control* (San Francisco: Holden Day, 1970). For a summary of their work see G. Box and G. Jenkins, 'Some Recent Advances in Forecasting and Control', *Journal of the Royal Statistical Society*, Series C, 17 (1968), 92–109.

Thus if $Z_0 = 50$, then $Z_1 = 49$ where $a_1 = -1$; $Z_2 = 48$ where $a_2 = -1$; and $Z_3 = 49$ where $a_3 = +1$ and so on. Note that the value of a_t is not influenced by the value of a_{t-1} and thus the stochastic disturbance terms a_t, a_{t-1} etc. are independent. In this example the observations fluctuate around a mean value of 50 since positive and negative deviations are equally likely. A series which does this is known as a stationary series. This can be distinguished from a non-stationary series which drifts upwards or downwards over time.

In this example a_t can take on only one of two possible values, but more generally the stochastic term can take on any one of a range of possible values. A more general version of (1) would allow this, assigning a probability distribution to the range of values taken on by a_t. It is also likely that the recent values of the stochastic term will have more impact than more remote values on the current observation of the series. Thus a series of weights which decline in value over time can be attached to the stochastic terms with varying lags to take this into account. A more general version of (1) can be written as follows:

$$Z_t = \mu + a_t + \theta_1 a_{t-1} + \theta_2 a_{t-2} + \dots + \theta_q a_{t-q}, \tag{2}$$

where μ is the mean value of the series, and $\theta_1 \dots \theta_q$ are weights which decline in value from 1 to q.

$$\begin{aligned} E(a_{t-i}\, a_{t-j}) &= 0 \text{ when } i \neq j \\ &\sigma_u^2 \text{ when } i = j. \end{aligned}$$

The last of these expressions makes the point that successive values of the disturbance terms are independent of one another and thus the expected value of their covariance is zero; this is the same as (1). It is also the case that the value of the disturbance term at any one point of time will fall within a range with a given probability. The variance of this probability distribution is measured by σ_u^2. This more general model is known as a moving-average model of order q, because the observations are a moving average of the stochastic terms reaching back over q periods.

If the process is stationary then its behaviour is not influenced by the point of time which it has reached. Thus

$$Z_{t-1} = \mu + a_{t-1} + \theta_1 a_{t-2} + \theta_2 a_{t-3} + \dots + \theta_{q-1} a_{t-q}. \tag{3}$$

If this is expressed in terms of a_{t-1}, it can be substituted into (2) giving:

$$Z_t = \mu + a_t + \theta_1 (Z_{t-1} - \mu - \theta_1 a_{t-2} \dots) + \theta_2 a_{t-2} + \dots + \theta_q a_{t-q}$$

$$Z_t = \mu(1-\theta_1) + \theta_1 Z_{t-1} + a_t - \theta_1^2 a_{t-2} \dots + \theta_2 a_{t-2} + \dots + \theta_q a_{t-q}.$$

Similarly a_{t-2}, a_{t-3} and successive stochastic terms may be eliminated from (3) by substitution, leaving an expression for Z_t of the form:

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \dots + \phi_p Z_{t-p} + \delta + a_t. \tag{4}$$

In this version, the current observation is expressed as a function of p past observations. The weights $\phi_1 \dots \phi_p$ are functions of the θ_i weights, and the constant term δ is a function of μ and the θ_i weights. Expression (4) is referred to as an autoregressive process of order p. This term 'autoregressive' comes from the fact that (4) is essentially a regression model in which variable Z_t is related to its own previous values rather than to a set of independent variables.

In general any linear, stochastic process may be written in the form of (2), as a weighted sum of current and past disturbances. When there is a finite number of non-zero terms, it is a moving-average process. The process may be alternatively written in the form of (4), as a weighted sum of all past observations, where again there is a finite number of non-zero terms. This is known as an autoregressive process.

THE BOX-JENKINS MODEL-BUILDING STRATEGY

The Box-Jenkins model-building strategy seeks to identify and estimate the most efficient stochastic model which fits a time series. The strategy involves four related and sequential stages: identifying a likely class of models, estimating the parameters of the most efficient and parsimonious model from this class, carrying out diagnostic checks on the model to test its efficiency, and finally, forecasting the future behaviour of the series from the model. Most time series can be modelled by a stochastic process of a low order. It is often possible to model a series by a first-order moving-average model of the following type:

$$z_t = \mu + a_t - \theta_1 a_{t-1} \tag{5}$$

where by convention the θ_1 weight has a negative sign. Other series may be efficiently modelled by a first-order, autoregressive process of the following type:

$$z_t = \delta + \phi Z_{t-1} + a_t. \tag{6}$$

Box and Jenkins particularly explored a class of mixed autoregressive moving-average models which have a wide variety of applications. A first-order autoregressive moving-average model is expressed as follows:

$$Z_t = \phi_1 Z_{t-1} + \delta + a_t - \theta_1 a_{t-1}. \tag{7}$$

The most appropriate model for a series depends on the behaviour of that series. A first-order moving-average model has a system memory of only one period, i.e. Z_t is not influenced by observations earlier than Z_{t-1}. In the first-order autoregressive model, on the other hand, current observations are influenced by a series of previous observations up to a given lag. The influence of the previous observations declines exponentially as the lags increase. These properties are illustrated in Figure 1, which contains forty observations of both a first-order autoregressive model and a first-order moving-average model.

The autoregressive process in Figure 1 'tracks' over time, i.e. when the series falls below zero it tends to stay there for a succession of time periods. This is because a series of observations are related to each other. The moving-average model in Figure 1 fluctuates above and below its mean value at random, because the series of observations are not related. A first-order autoregressive moving-average model combines the properties of both. In this case a series of observations are related over

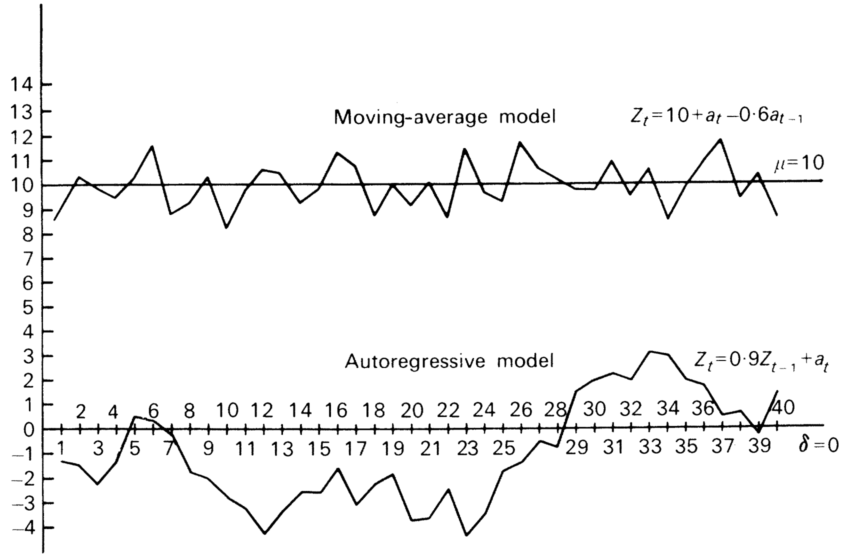

Fig. 1. A simulated first-order autoregressive process and a moving average process.

time, but the process exhibits more complex fluctuations than a purely autoregressive model.

In a first-order autoregressive model the influence of previous observations on the current observation is measured by the autoregressive coefficient ϕ. A model with a relatively small coefficient might not track nearly so clearly as the model in Figure 1. In fact when the relationship between successive observations is weak it might be more efficient to model the series as a moving-average process. An autoregressive process with a negative coefficient will shift sequentially between values above the mean and values below the mean, giving a very spiky appearance. Autoregressive, or moving-average models of a higher order, can behave in even more complex ways. Thus it is generally not possible to identify a series merely by inspection, although this helps. The main tool of analysis at the identification stage of the procedure is the autocorrelation function.

The autocorrelation function consists of a series of correlation coefficients which relate all observations up to the current one with all observations up to the current, lagged one or more periods. Thus the autocorrelation coefficient lagged one period is given by:

$$r_1 = \sum_{t=2}^{N} \frac{(Z_t - \bar{Z})(Z_{t-1} - \bar{Z})}{\sum_{t=1}^{N} (Z_t - \bar{Z})}, \tag{8}$$

where $\bar{Z}$ is the series mean.

Autocorrelation coefficients can be calculated for a succession of lags, and since the value of the coefficient is a function of the lag the set of coefficients is referred to as the autocorrelation function.

The behaviour of the autocorrelation function calculated for a particular series assists in identifying the series. Thus if the only non-zero coefficient of the function is the first, this strongly suggests that the series is a first-order moving-average process. This is because such a process has a system memory of only one period and hence we should not expect a series of observations to be related. On the other hand, if the autocorrelation function has values which decline exponentially for increasing lags, this would suggest that the series is a first-order autoregressive process. A first-order autoregressive moving-average process has an autocorrelation function which behaves in a similar way to the autoregressive model except the coefficients exhibit a tendency to oscillate over successive lags.

Another tool of analysis at the identification stage is the partial autocorrelation function, which particularly helps the identification of higher-order processes. For a process of a given order the partial autocorrelation function consists of a series of coefficients which measure the relationship between successive autocorrelation coefficients.[4] With higher-order autoregressive processes a series of autocorrelations will have associations between them which the partial autocorrelation function will pick up.

It has been suggested that some series might be modelled differently, particularly when they are complex. In the Box–Jenkins approach the most parsimonious model, i.e. the one with fewest parameters, is taken as the most efficient. But it may be useful on occasions to take a less parsimonious model, particularly if the accuracy of its forecasts is greater.

When identifying the model of a time series, one of the first things to determine is whether or not the process is stationary. This is not always clear. In general, if the autocorrelation function has significant coefficients at long lags this would suggest that the series is non-stationary. The reason for this can be seen by considering a non-stationary series which drifts upwards over time. In this case nearly all changes in observations after a given time will be positive. Thus observations up to t will correlate positively with observations up to $t+j$, where j is relatively large.

In a non-stationary model, the parameters of the system change over time because the system does not behave in the same way at different points of time. To model this directly would involve incorporating a series of extra parameters to measure the stage which the process had reached. It is far more efficient to transform the non-stationary model into a stationary model by 'differencing', that is, subtracting the previous from the current observations, as follows:

$$Z_t^* = Z_t - Z_{t-1}.$$

Some complex series may require differencing more than once to induce stationarity. It is important, however, not to difference a series unnecessarily, since this can produce a highly complex model which is an artifact of the differencing

[4] For example, in the case of a first-order autoregressive process

$$r_2 = \phi_{12} r_1$$

where r_1 and r_2 are the autocorrelation coefficients between observations up to t and observations up to $t-1$ and $t-2$ respectively. ϕ_{12} is the partial autocorrelation coefficient. See G. Box and G. Jenkins, 'Time Series Analysis', pp. 64–5.

procedure. The most parsimonious model is the one which involves least differencing.

Having tentatively identified the model the next stage is to estimate it. The estimation procedure is a non-linear least squares procedure. Thus the parameter estimates minimize the square of the residuals of the model, in the same way as linear regression. However, the fact that the model is non-linear makes the estimation process rather more complex than ordinary least squares regression.[5]

The diagnostic checking stage of the procedure involves testing the model's goodness of fit, by adding parameters and also examining residuals. Thus, in the case of a first-order autoregressive model one might add a moving-average parameter or a second autoregressive parameter and test whether the new parameter is statistically significant. If new parameters are not significant this suggests that the original model is adequate. Another check involves the residuals. If the model fits adequately the residuals should be a random uncorrelated series of observations, which are referred to as 'white noise'. But if the residuals reveal some systematic pattern it follows that the model is failing to pick up this pattern. The residual autocorrelation function can be compared with a white noise series using a χ^2 goodness-of-fit test. If the χ^2 statistic is significant at the usual levels that means the residuals differ from white noise, and the model is inadequate.

This strategy makes it possible to identify the most efficient model of a series, which can then be used for forecasting. Although Box–Jenkins models have rarely been used in the social sciences outside economics,[6] it has been shown that they are more accurate than any rivals in one-step-ahead forecasting.[7] As the following two sections show, they provide an efficient technique for forecasting the vote shares of the major parties, and indirectly their seat shares, at a general election on the basis of an opinion-poll time series.

[5] From (7) it can be seen that

$$a_t = Z_t - \phi_1 Z_{t-1} - \delta + \theta_1 a_{t-1}.$$

We seek to find values of ϕ and θ which minimize Σa_t^2. But to obtain a_t it is necessary to know a_{t-1}, which is not available. Thus the procedure involves assigning starting values to a_{t-1} (usually zero), and then calculating the sum of squared residuals. The lagged value of the residuals can then be used in a subsequent estimation to obtain a better fit. The procedure is repeated iteratively until the parameter values converge on stable values. For a full discussion of this see O. N. Anderson, *Time Series Analysis and Forecasting: The Box–Jenkins Approach* (London and Boston: Butterworths, 1976), Chap. 8.

[6] The one application of Box–Jenkins modelling to political science is by Hibbs. Using British unemployment data for the period 1948–75 he assessed the effects of Government policy interventions on the level of unemployment. See D. A. Hibbs, 'On Analyzing the Effects of Policy Interventions: Box–Jenkins and Box–Tiao versus Structural Equation Models' in D. Heise, ed., *Sociological Methodology* (San Francisco and Washington: Jossey-Bass, 1977), pp. 137–79. See also D. A. Hibbs, 'Political Parties and Macroeconomic Policy', *American Political Science Review*, LXXI (1977), 1467–87, where he examines United States' unemployment data as well as the United Kingdom data.

[7] D. J. Reid, 'A Comparison of Forecasting Techniques on Economic Time Series', in M. J. Bramson, ed., *Forecasting in Action* (London: Operational Research Society, 1972). Reid used a set of 113 time series, consisting mainly of economic variables, and examined the comparative efficiency of different forecasting techniques.

TIME SERIES MODELS OF POLL DATA

The Box–Jenkins model-building strategy was applied to four different series of Gallup poll data for the United Kingdom. The series were: support for the Labour party, support for the Conservative party, government popularity which was derived from these two, and finally prime minister's popularity. The series were measured from January 1947 to December 1975, giving 348 monthly observations altogether.[8] Table 1 contains descriptive statistics for each of the series.

It can be seen from Table 1 that on average the Labour party was marginally more popular than the Conservative party over this period. The standard deviations of the series measure the extent to which they fluctuate over time, and this is about the same for both series. The government popularity series shows that on average the opposition party is 3·6 per cent more popular than the governing party. Moreover, this series fluctuates considerably more than the party popularity series from which it is derived. Of all the series prime minister's popularity fluctuates most, although the mean level of this series is relatively high compared with party popularity.

TABLE 1 *The Means and Standard Deviations of the Poll Data*

	Labour popularity	Conservative popularity	Government popularity	Prime minister's popularity
Mean	44·08	43·12	−3·61	47·71
Standard deviation	4·72	4·70	7·55	9·88
N	348	348	348	348

The sample autocorrelation functions for each series lagged up to twenty-five periods appears in Table 2. The system memory of each series can be determined from this table, i.e. the period of time over which current opinion is influenced by previous opinion. Bartlett's test of statistical significance[9] can be applied to successive coefficients of the autocorrelation function until one arrives at a coefficient which is no longer statistically significant at the usual levels. This determines the system memory. It can be seen from Table 2 that the system memories of the Labour, Conservative and 'prime minister's' popularity series are

[8] The data were obtained from various issues of the *Gallup Political Index* (London: Gallup Poll Ltd.) except for some of the earlier post-war data which was kindly supplied by Mr C. A. Goodhart. The data for the parties represent the percentage of respondents who chose that particular party in response to the question: 'If there were a General Election tomorrow, how would you vote?' The series used was Series C in the *Gallup Political Index*, i.e. the observations with 'don't knows' excluded, since these provide a better fit for electoral forecasting. The data on 'prime minister's popularity' refers to the percentage of respondents satisfied with the job Mr X is doing as prime minister. 'Government popularity' refers to the percentage support for the incumbent party minus the percentage support for the opposition party, using the above question, and again excluding the 'don't knows'.

[9] See Box and Jenkins, 'Time Series Analysis', pp. 34–6.

approximately the same at just over one year. The memory of the government popularity series is significantly shorter at about ten months. This is accounted for by the asymmetry in government and opposition popularity noted by Miller and Mackie.[10] Since government popularity is not the mirror image of opposition popularity, this series fluctuates more than the two series which it is derived from, and also has a shorter memory.

TABLE 2 *The Autocorrelation Functions Lagged Up to Twenty-five Periods*

Lag	Labour popularity	Conservative popularity	Government popularity	Prime minister's popularity
1	·86	·84	·83	·91
2	·79	·77	·73	·84
3	·73	·70	·64	·79
4	·67	·65	·56	·75
5	·64	·63	·52	·72
6	·61	·61	·47	·69
7	·60	·59	·45	·66
8	·55	·59	·40	·60
9	·49	·52	·33	·56
10	·44	·48	·29*	·53
11	·40	·45	·27	·51
12	·37	·43	·24	·49
13	·34*	·39	·18	·46
14	·30	·36*	·14	·42*
15	·26	·32	·10	·38
16	·24	·29	·03	·34
17	·19	·25	−·01	·30
18	·14	·21	−·06	·28
19	·12	·20	−·09	·24
20	·08	·18	−·15	·20
21	·04	·15	−·19	·16
22	−·02	·14	−·19	·13
23	−·06	·09	−·23	·10
24	−·11	·06	−·26	·06
25	−·13	·06	−·27	·04

* Autocorrelations up to the asterisked lag are significant at the ·05 level.

The autocorrelation functions rule out purely moving-average models, because of these system memories. Goodhart and Bhansali suggested that the poll series could be adequately modelled by a first-order autoregressive process.[11] This specification was accordingly tried first for each of the series. However, the resulting models failed to pass the diagnostic checks. Either the residuals were not white noise, or

[10] Miller and Mackie, 'The Electoral Cycle', p. 274.

[11] Goodhart and Bhansali, 'Political Economy', p. 92.

parameters added to the models proved statistically significant, or both. The partial autocorrelation functions of each series, which appear in Table 3, confirm this. They do not cut off after the first lag, and contain small but significant coefficients at longer lags. This is not the behaviour characteristic of a first-order autoregressive process.

TABLE 3 *The Partial Autocorrelation Functions*

Lag	Labour popularity	Conservative popularity	Government popularity	Prime minister's popularity
1	·86*	·84*	·83*	·91*
2	·19*	·24*	·16*	·08
3	·03	·02	−·03	·11*
4	·03	·08	·03	·07
5	·08	·10	·06	·03
6	·05	·08	·03	·00
7	·10	·04	·05	·02
8	−·11*	·08	−·03	−·12*
9	−·10	−·17*	·12*	−·01
10	−·06	−·01	·01	·02

* Significant at the ·05 level.

As a result, all the models were re-estimated in the forms of first-order autoregressive moving-average models. This specification proved adequate for all of the series. The estimates of the different models appear in Table 4 where the standard errors of the coefficients appear in parentheses, and the χ^2 statistics refer to the goodness of fit of the residual autocorrelation functions to a white noise process. None of the residual series significantly differ from a white noise process, which indicates that the models fit adequately. One small qualification to this is the Conservative series, which comes closer than the others to failing the diagnostic test. The autocorrelation function for the residual series was examined closely. Any model which fits adequately should have a residual autocorrelation function with no significant coefficients. In the case of the Conservative popularity series the residual autocorrelation function contains a statistically significant component at a lag of eight months; such a coefficient is consistent with the presence of a short-term cycle of popularity in the Conservative series. Attempts to investigate such a cycle proved fruitless, however; this coefficient can therefore be regarded as due to chance. But it does explain the poorer fit of the Conservative model.

The main characteristic of Table 4 is the remarkable similarity between the models for each of the series. In each case the current observation is very highly related to the previous observation of the poll. However, to predict the current observation efficiently from the previous observation two stochastic disturbance terms are required: one associated with the current observation and a second associated with the previous observation, weighted so that it is much less important than the current

TABLE 4 *The Autoregressive Moving-Average Models of the Poll Series*

Labour popularity	
$Y_t = \cdot 92\ Y_{t-1} + a_t - \cdot 23\ a_{t-1}$	
$(\cdot 02)\qquad(\cdot 05)$	
Residual variance = 5·44	χ^2 residuals = 24·98 with 18 D.F.
$N = 348\quad R^2 = \cdot 76$	
Conservative popularity	
$Z_t = \cdot 92\ Z_{t-1} + a_t - \cdot 30\ a_{t-1}$	
$(\cdot 02)\qquad(\cdot 05)$	
Residual variance = 6·26	χ^2 residuals = 33·3 with 23 D.F.
$N = 348\quad R^2 = \cdot 72$	
Government popularity	
$g_t = \cdot 88\ g_{t-1} + a_t - \cdot 18\ a_{t-1}$	
$(\cdot 02)\qquad(\cdot 05)$	
Residual variance = 17·59	χ^2 residuals = 24·80 with 23 D.F.
$N = 348\quad R^2 = \cdot 69$	
Prime minister's popularity	
$p_t = \cdot 93\ p_{t-1} + a_t - \cdot 11\ a_{t-1}$	
$(\cdot 02)\qquad(\cdot 05)$	
Residual variance = 17·05	χ^2 residuals = 25·07 with 23 D.F.
$N = 348\quad R^2 = \cdot 83$	

disturbance term. It appears that the four aspects of public opinion studied in this paper behave in a very similar way over time. They have a broadly similar memory and are subject to similar stochastic influences.

Having identified and estimated the models, the next step is to forecast from the models. In the case of a first-order autoregressive moving-average model, the one-step-ahead forecasting equation is given by:

$$Z_t(1) = \phi Z_t + \delta + a_t - \theta a_{t-1}. \tag{9}$$

Thus the current forecast consists of a constant term (δ), which we saw earlier is related to the series mean, an autoregressive relationship with the current observation (ϕ), a disturbance term associated with the current observation (a_t) and a disturbance term related to the previous observation (a_{t-1}), which is weighted (θ).[12]

It is necessary to determine the accuracy of these forecasting models over a reasonably large sample of observations. The models were used to forecast the last fifty observations of the series in each case, and some diagnostic statistics associated with these forecasts appear in Table 5.

The mean square error statistic in Table 5 is calculated from the following expression

$$\text{M.S.E.} = \sum_{t=1}^{50} (Z_t - \hat{Z}_t)^2 \frac{1}{50},$$

[12] The parameters are estimated from the observations expressed in deviations from the mean. In this case the constant term is zero when the series is stationary, and hence is omitted from the models in Table 4.

Z_t is the observation at time t,

$\hat{Z}_t$ is the forecast at time t.

The M.S.E. is the standard goodness-of-fit measure in forecasting. In Table 5 the Labour and Conservative models forecast the observations most closely. The large M.S.E. for the government popularity series suggests that this is unsuitable for forecasting. It is much better to forecast the party support, rather than to combine party popularities into a government popularity series.

The period of time over which the forecasts are made provides a fairly stringent test of their efficiency. During this period there were two general elections, one held in an atmosphere of constitutional crisis. Opinion fluctuated more widely at this time than at any other in post-war Britain.

TABLE 5 *Diagnostics on the Forecasts of the Last Fifty Observations*

	Labour popularity	Conservative popularity	Government popularity	Prime minister's popularity
Mean square error of forecast	8·01	7·56	26·74	12·83
Average percent forecast error	2·24	2·03	3·86	2·84

The M.S.E. statistics are not always intuitively interpretable, so the average percentage forecast error is included in Table 5. The average error for the Conservative series was just above 2 per cent, and for the Labour series about 2·25 per cent. The size of the average error or the M.S.E. before forecasting becomes inefficient is arbitrary, but clearly once the average error approaches 4 per cent, as in the case of government popularity, the accuracy of forecasts becomes rather poor.

To illustrate how the forecast series behaves in relation to the original series, the last fifty observations and the forecasts of the Labour series appear in Figure 2. It can be seen that the forecasts track the observations fairly closely and are only seriously in error when sharp discontinuities exist in the observations, as in the case of the surge in Labour popularity after the February 1974 general election. An additional test of the efficiency of the forecasts is obtained by comparing the error series with a white noise process in the same way as described earlier. This was done for each of the models, and in each case the error series was indistinguishable from white noise.

It is worth examining the efficiency of the Box–Jenkins forecasts for the nine general elections which took place during the period of the observations. Of course it is more difficult to make inferences from a sample of only nine cases than it is with fifty observations, but the results are interesting. Table 6 contains the forecasts of Labour's percentage share of the total poll made from all observations up to the last poll available before the election campaign began. Each forecast is one of a series made from a starting point six months prior to the election month. Ninety per cent

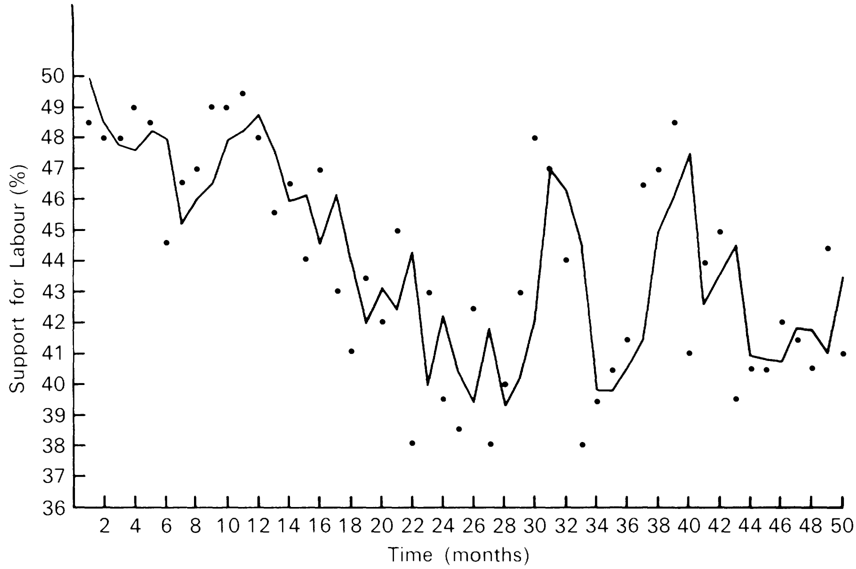

Fig. 2. One-step-ahead forecasts of Labour popularity, October 1971 to December 1975.

TABLE 6 *Forecasts of Labour Support in General Elections 1950–74*

Election year	Forecasts from observations up to the last pre-campaign poll: Lower*	Forecast	Upper*	Labour's actual % share of poll	$\|e_i\|$ error or residual†
1950	38·8	42·7	46·5	46·1	3·4
1951	36·5	40·4	44·2	48·7	8·3
1955	42·1	45·9	49·8	46·3	0·4
1959	39·3	43·2	47·0	43·8	0·6
1964	43·0	46·8	50·7	44·1	2·7
1966	44·2	48·1	51·9	47·9	0·2
1970	43·0	47·1	50·9	43·0	4·1
1974 (Feb.)	35·5	39·4	43·2	37·1	0·3
1974 (Oct.)	36·8	40·7	44·5	39·3	1·4

* Denotes the lower and upper bounds of the 90 per cent confidence interval of the forecast.
† Mean of items in this column, 2·3; mean excluding 1951, 1·6.

confidence intervals are included; thus the forecaster can be 90 per cent confident that the true outcome will be within the interval defined.

Obviously the 90 per cent confidence interval will be wrong on average one time in ten, and this is in fact what happens in the case of the 1951 forecast. The large inaccuracy of the 1951 forecast is not, in fact, due to the model so much as the Gallup data, which were extremely inaccurate in that year.[13] If 1951 is treated as an exception, the mean error of the forecast is 1·6 per cent for the remaining eight general elections.

From this small sample of elections it is possible to make inference about the mean error of the forecast for the population of all possible general elections. If $\bar{X}$ and S^2 denote the mean and the variance respectively of this (assumed) random sample of forecasting errors, and if it is assumed that the population of all forecasting errors is normally distributed with mean μ and variance σ^2, then the statistic

$$\bar{X}-\mu \Big/ \frac{S}{\sqrt{n}}$$

has a t distribution with $n-1$ degrees of freedom.[14] A 90 per cent confidence interval estimate of μ is given by:

$$\mu = \bar{X} \pm t_{\cdot 10}\left(\frac{S}{\sqrt{n}}\right),$$

which in this case is:

$$\mu = 2{\cdot}3 \pm 1{\cdot}64.$$

Thus there is a one in twenty probability that the mean forecasting error for all possible general elections will exceed approximately 4 per cent. If the sample were larger it would be possible to estimate this more accurately.

One noticeable characteristic of the different poll series is that they all appear to contain cycles. Popularity cycles for the United Kingdom were examined by Goodhart and Bhansali using spectral analysis.[15] Goodhart and Bhansali pointed out that 'there may not be regular major cycles in these political popularity series, and that contrary to popular belief there may not be any consistent seasonal patterns.'[16] The Box–Jenkins models can be used to estimate regularly recurring cycles such as seasonal variations, but not irregular cycles. For short-term forecasting, irregular cycles are not a critical issue anyway, unless they are both short-term and highly significant. Only if one sought to forecast many months into the future would cyclical components have to be incorporated, and this would have to be done on an *ad hoc* basis if the cycles were irregular.

Another important point is that models incorporating autoregressive components will very often exhibit irregular cycles. As we observed earlier, an autoregressive process with a high positive coefficient 'tracks' over time and this produces irregular

[13] This was inferred from the residuals of the forecasting model. See Table 6, p. 231.

[14] See J. E. Freund, *Mathematical Statistics* (London: Prentice-Hall International, 1972), pp. 220–3.

[15] Goodhart and Bhansali, 'Political Economy', pp. 94–106. Spectral analysis involves decomposing the series into basic frequencies. If the series is stationary it can be decomposed into a set of uncorrelated components, each of which is related to a particular frequency in the data. It is then possible to identify the contribution of each frequency to the total variation in the series. The procedure is analogous to the analysis of variance. The periodicity of significant cycles can be identified using this technique. See M. G. Kendall, *Time Series* (London: C. Griffin, 1973), Chap. 8.

[16] Goodhart and Bhansali, 'Political Economy', p. 97.

cycles. The system stays above the mean for several periods until a series of shocks moves it below the mean, and in turn it will stay there for several periods. Such irregular cycles are caused by the random shocks, rather than by any learning or adaptive behaviour. In the context of the opinion polls, these random shocks are the unique events of political life: a key strike, a financial scandal or a political row. They are of great interest to the historian, since in retrospect they may turn out to be the decisive moments of history. But they are inherently unpredictable, and must be regarded by the forecaster as a random shock to the system.

One interesting aspect to the influence of these political events can be inferred from the estimated models. A particular event, such as a key political speech which attracts wide publicity, will influence public opinion directly for two months, via a_t and a_{t-1}. Subsequent influences on opinion are indirect via the autoregressive relationship. In other words, the influence of any event may be felt in the system for several months just because of the inertia of public opinion. In the absence of any further shocks an event will finally cease to influence opinion after just over a year. Since random shocks are occurring the whole time the longevity of a particular event may be much shorter than this, because it is overtaken by other events. Moreover, these random shocks must be distinguished from other random influences on the polls, such as measurement error and sampling variability, which themselves create irregularities.

The main theme of this paper has been the forecasting of poll movements. However, outcomes need to be forecast in terms of parliamentary seats. This involves additional problems, and is discussed next.

FORECASTING ELECTORAL OUTCOMES

There are two major difficulties to the forecasting of electoral outcomes from models of the poll series. The first is that the election itself influences the polls, and the forecast is made on the assumption that the underlying model of public opinion is unchanged. The second is the problem of inferring the number of seats which a party will capture with a given share of the total vote.

As regards the first problem, the election can be regarded as a random shock that influences the system but does not disturb the underlying structure. The efficiency of the forecast might then be reduced, but it would still be a useful guide. On the other hand, if the underlying structure changes in form, or there are significant changes in the parameters of the models at election times, this would invalidate any attempt to forecast. These possibilities can be examined by comparing the residuals of the model applied to election results with the residuals of the model applied at other times. If the residuals from the election results are significantly larger than at other periods this would suggest that the models do not adequately fit at election times.

We assume that the residuals obtained from fitting the model of Labour support to all the observations are normally distributed, with a zero mean and constant variance. The diagnostic statistics do in fact support this assumption. Because of this assumption it is possible to calculate the probability that a given residual will exceed a specified size. If the distribution of residuals is standardized then:

The probability that $\frac{|e_i|-\bar{X}}{S} > 1{\cdot}96$ is $\leqslant {\cdot}05$.

Where $|e_i|$ is the absolute value of the i'th residual,

$\bar{X}$ is the mean of the residual distribution ($=0$),

S is the standard deviation of the residual distribution ($=\sqrt{5{\cdot}44}$ from Table 4).

Thus the probability of $|e_i| > 4{\cdot}57$ is $\leqslant {\cdot}05$.

It can be seen in Table 6 (p. 231) that only one residual from the forecast model exceeds a value of 4·57. This therefore suggests that the residuals of the model applied to electoral outcomes do not significantly differ from the residuals at other times, and the models are adequate in forecasting elections.

The second problem – calculating the share of parliamentary seats from the share of the vote – has been examined by a number of authors.[17] One means for translating the share of the vote into shares of seats is known as the cubic rule. This asserts that the number of seats obtained by a party is approximately equal to the cube of its share of the vote. This can be written:

$$S_i = \frac{V_i^3}{\sum_{j=1}^{n} V_j^3}, \tag{10}$$

S_i is the share of the seats obtained by party i,

V_i is the share of the vote obtained by party i,

n is the number of parties.

Kendall and Stuart examined the statistical properties of this relationship and found that it was based on the fact that the proportion of the total vote obtained by a party in successive British general elections was approximately a normal distribution.[18] This proportion will vary in successive general elections, but the variance of the distribution of proportions would not. It turns out that the variance of the actual distribution of seat shares is approximately the same as the variance of the distribution of seat shares predicted by the cube rule. This fact accounts for the reliability of the relationship.

More recent work with British elections by Laakso[19] has shown that for the post-war period the distribution of seat shares is predicted rather better by a 'two-and-a-half' than by a cube rule. In other words the power terms in expression (10) should be 2·5 not 3·0. Accordingly this two-and-a-half rule is used to forecast the share of the seats from the forecast share of the votes. This is done in Table 7.

[17] See M. G. Kendall and A. Stuart, 'The Law of Cubic Proportions in Electoral Results', *British Journal of Sociology*, I (1950), 183–97; J. G. March, 'Party Legislative Representation as a Function of Election Results', *Public Opinion Quarterly*, XI (1957), 521–42; T. Qualter, 'Seats and Votes: An Application of the Cube Law to the Canadian Electoral System', *Canadian Journal of Political Science*, I (1968), 336–44; and E. R. Tufte, 'The Relationship between Seats and Votes in Two-party Systems', *American Political Science Review*, LXVII (1973), 540–54.

[18] Kendall and Stuart, 'The Law of Cubic Proportions', p. 190.

[19] Markku Laakso, 'Should a Two-and-a-Half Law replace the Cube Law in British Elections?', *British Journal of Political Science* (forthcoming).

TABLE 7 *The Prediction of Seat Shares from the Actual and Forecast Vote Shares of the Two Major Parties for Labour 1950–74*

Election year	Actual % share of seats	2½ rule prediction of share from vote	No. of seats in error*	2½ rule prediction of share from forecast	No. of seats in error from forecast†
1950	51·4	53·8	+14	48·3	−19
1951	47·9	50·9	+19	36·1	−74
1955	44·5	45·5	+6	48·8	+27
1959	41·4	42·5	+7	42·6	+8
1964	51·0	50·9	−1	53·9	+18
1966	58·9	58·4	−3	57·4	−9
1970	46·5	45·2	−8	49·7	+20
1974	50·3	48·5	−11	50·3	0
1974	53·6	55·7	+13	55·0	+9

* |mean| = 9·1.
† |mean| = 20·4; |median| = 18·0.

The two-and-a-half rule is very inaccurate at predicting the share of seats going to third parties like the Liberals. It does not allow for the bias against third parties inherent in the first-past-the-post electoral system. Thus the calculations in Table 7 are based on forecasts of the Labour and Conservative vote only.

It can be seen from Table 7 that the average (absolute) error in forecasting the Labour share of the seats held by the two major parties is just over twenty seats. If 1951 is excluded it is approximately fourteen seats. This figure includes error due to the imperfect fit of the two-and-a-half rule, which accounts for 45 per cent of the total error. Thus it can be inferred that the net absolute error of the forecasting model for these nine general elections is approximately eleven seats.

THE FORECASTING MODEL AND A SUMMER 1978 GENERAL ELECTION

In May and June 1978 there was considerable press speculation about the possibility of a summer general election. Labour had recovered much of its popularity in the opinion polls, and the Hamilton by-election result suggested that Labour had checked the growth of Nationalist support in Scotland.[20] Even though the ill-fated snap election of 1970 provided a bad precedent, it was thought that the Prime Minister might try an early election before the summer holidays.

On the basis of Gallup poll data up to and including May 1978 the forecasting model was used to predict the Labour and Conservative vote shares for hypothetical

[20] In the Hamilton by-election on the 31 May 1978 there was a 2·5 per cent swing to Labour from the Scottish National party. The Gallup poll for Great Britain taken on the 18 May 1978 showed Labour and the Conservatives to be neck-and-neck in the polls, each with a support of 43·5 per cent. This contrasted sharply with the position in May 1977 when the Conservatives had a 20·5 per cent lead over Labour. See *The Gallup Political Index*, Report No. 214 (London: Gallup Polls Ltd., 1978), Table 1.

general elections in June and July. The two-and-a-half rule was then used to predict the share of the seats from the forecast vote shares. As we have seen, the two-and-a-half rule predicts minor party seat shares very inaccurately. It was therefore necessary to make two different assumptions about minor party seat shares: first, that the two major parties would capture the same percentage of the seats as they did in 1970 (97·9 per cent), and second, that they would capture the same percentage as they did in October, 1974 (93·7 per cent).

TABLE 8 *The Estimates of Vote and Seat Shares in Hypothetical General Elections, June and July, 1978*

		June	July
Vote shares	Labour	43·4	43·0
	Conservative	44·2	45·6
Seats (1)*	Labour	304	285
	Conservatives	318	331
Seats (2)†	Labour	290	275
	Conservatives	304	319

* Assumes the share of seats won by the two major parties combined is the same as 1970.
† Assumes the share of seats won by the two major parties combined is the same as October 1974.

The results of this forecast appear in Table 8 and show that on the 1970 assumption the Conservatives would have obtained an overall majority of one in June, and of fourteen in July. On 1974 assumptions there would have been no overall majority in June, but a Conservative majority of two in July.

In view of the forecasting errors discussed earlier the probability of a coalition government resulting from no overall majority for any party was quite significant. At all events the forecasting results show that Labour would have been unwise to call a summer election.

CONCLUSIONS

The Box–Jenkins model-building strategy makes it possible to develop efficient models of British public opinion for purposes of forecasting. The models are most efficient when applied to the Labour and Conservative popularity series. The government popularity series is not really efficient for purposes of forecasting due to the lack of any strong inverse relationship between government and opposition popularity. When the Labour support model is used to forecast the share of the poll for the nine general elections since 1950 the absolute mean error is 2·3 per cent. When the models are used to forecast the Labour share of the two party seats, the average absolute error is approximately twenty seats. But a significant proportion of this error is due to inaccuracies in the transformation used to relate vote shares to seat shares.